



SmartExam

FYP-1 Quality Assurance & Testing Report

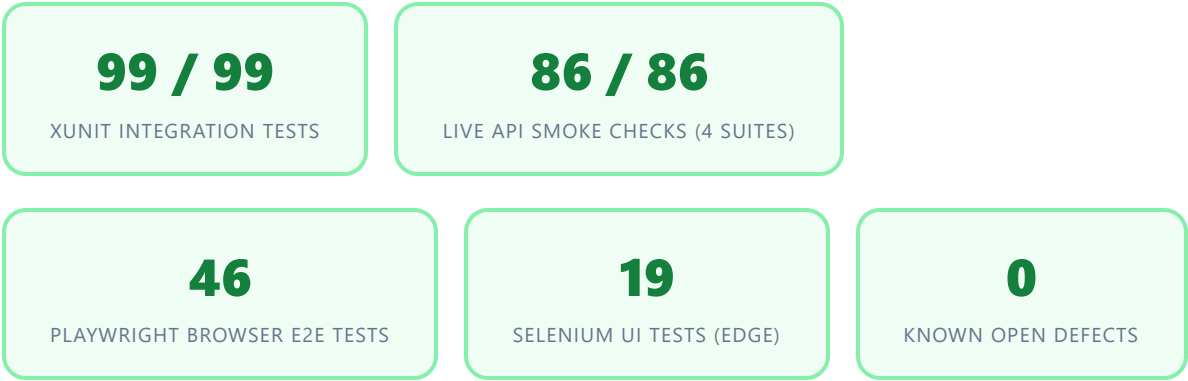
Unit · Integration · End-to-End · Security / Negative Testing

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Document 2 of 3 · Generated July 2026 · All figures from actual executed runs

1. Results at a Glance



Every feature shipped in FYP-1 is covered by at least two independent test layers, and every endpoint has explicit **negative-path** tests (wrong role, invalid input, conflicting state, and attack attempts such as IDOR).

2. Test Strategy — Four Layers

LAYER	TOOLING	WHAT IT PROVES
1. Integration tests	xUnit + WebApplicationFactory (in-memory DB, real HTTP pipeline incl. JWT middleware)	Every controller behaves correctly through the full ASP.NET pipeline — auth, validation, business rules, persistence
2. Live API smoke	PowerShell suites against the running backend + real Neon PostgreSQL	Behavior holds on the production database (migrations, FK constraints, real enum/JSON serialization)
3. Browser E2E	Playwright (46 tests) + Selenium/Edge (19 tests)	The React panel works end-to-end from a real browser: login flows, role redirects, exam authoring, monitoring pages
4. Manual scenario runs	Scripted demo walkthroughs (see Document 3)	Cross-application flows: WPF client ↔ API ↔ teacher dashboard in real time

3. xUnit Integration Suite (99 tests, 100% pass)

AREA	TESTS	COVERAGE HIGHLIGHTS
Authentication & HWID	9	Valid/invalid logins, student HWID bind + mismatch + missing-HWID, /me, logout
Users & Roles (M7) + Auth hardening (M1)	25	CRUD + guards, role filters, CSV import, Invigilator role, audit log writes, revoked-token rejection, deactivation enforcement, change-password round-trip
Exams (M8 + M9 + M10)	45	Exam lifecycle, MCQ validation + exact grading, answer-key stripping for students, edit/delete guards, seating conflicts, roster add/remove, extend-time bounds, heartbeat clock, monitoring events, violation persistence
Notifications	7	Send/receive, mark-read, exam announcements, role guards

Analytics & Reports	6	Exam summary KPIs, system analytics, PDF report generation + path-traversal protection
Student Portal	6	Dashboard stats, exam list, results, violations visibility
Smoke	1	Bootstrap sanity

4. Live Smoke Suites vs Neon PostgreSQL (86 checks, 100% pass)

SUITE	CHECKS	SCOPE
Module 8 — Exam authoring	16 / 16	Create w/ MCQ + instructions → verify → edit → verify → delete; answer-key protection; student /current carries instructions + real proctor
Modules 1 + 7 — Auth & users	21 / 21	Audit rows, change-password, logout revocation, HWID hash + mismatch audit, Invigilator, edit/deactivate/reactivate/delete guards, batch import
Modules 9 + 10 — Seating & proctoring	18 / 18	Lab/workstation CRUD, seat assign + conflict, roster add/remove, extend-time, audit rows for every action
Full negative regression	31 / 31	All bug-fix verifications + the negative catalog below

5. Negative & Security Testing Catalog

Every check below **passes** — the system rejects the request with the exact status shown.

Authentication & Session (Module 1)

ATTACK / INVALID ACTION	EXPECTED	RESULT
Login with wrong password / unknown email	401	PASS
Student login from a different (unbound) device	401 DEVICE_MISMATCH	PASS
Student login without a hardware fingerprint	400	PASS
Using a token after logout (revoked JTI)	401	PASS
Using a token after admin deactivation / force-logout	401	PASS
Change password with wrong current password	400	PASS
Non-admin reading the audit log	403	PASS

User Management (Module 7)

ATTACK / INVALID ACTION	EXPECTED	RESULT
Student/teacher creating users or importing CSV	403	PASS

Editing a user to an email already in use	409	PASS
Admin deleting their own account / changing own role	400	PASS
Deleting a teacher who owns course sections	409	PASS
Deleting a student who has exam history	409	PASS

Exam Authoring (Module 8)

ATTACK / INVALID ACTION	EXPECTED	RESULT
Exam with no questions / marks ≤ 0 / blank question text	400	PASS
MCQ with < 2 options, empty options, or out-of-range answer key	400	PASS
MCQ carrying test cases / coding question with no test cases	400	PASS
Plagiarism threshold outside 0–100	400	PASS
Student creating / editing / deleting an exam	403	PASS
Editing or deleting an Active/Ended exam	409	PASS
Student fetching an exam: hidden test cases, MCQ keys, reference answers	stripped	PASS

Seating & Access Control (Module 9)

ATTACK / INVALID ACTION	EXPECTED	RESULT
Assigning a seat already taken in the same exam	409	PASS
Duplicate workstation machine-number in a lab	409	PASS
Adding the same student to an exam twice / adding a non-student	409 / 400	PASS
Removing a student who already has an exam session	409	PASS
Student assigning seats or creating labs	403	PASS

Live Proctoring & Telemetry (Module 10)

ATTACK / INVALID ACTION	EXPECTED	RESULT
Extend-time with 0 / negative / > 180 minutes, or on an ended exam	400 / 409	PASS
Student calling extend-time / force-submit	403	PASS
IDOR: student posting telemetry/violations into another student's session	404	PASS
Student joining a proctoring broadcast group (SignalR)	rejected	PASS
Teacher force-submitting a session from another teacher's section	rejected	PASS

6. Defects Found & Fixed During QA

DEFECT	ROOT CAUSE → FIX
Heartbeat/answer saves returned 500	FK violation on empty session id + an EF shadow foreign key (ExamSessionSessionId) → guarded inserts + explicit relationship mapping + migration
Teacher instructions never reached students	Wizard collected the field but dropped it from the payload → persisted end-to-end (Exam.Instructions)
Start Exam could never enable	Attendance/identity gating had no UI → pre-exam checklist added to the desktop client
Deactivate/force-logout didn't log the student out	Tokens were revoked server-side but never re-checked → per-request revocation middleware + client session-watch
Wrong exam shown to students with several assignments	Unordered FirstOrDefault → nearest-open-exam selection (dashboard + start kept consistent)
Exam edit crashed (500)	EF graph-fixup marked re-added children Modified during collection replacement → materialize + explicit AddRange (regression-tested)
Sensitive leaks (password hashes, answer keys, plagiarism telemetry to all clients)	Safe projections; per-exam SignalR groups; role/ownership checks on hub methods

Reproducibility: run `dotnet test` in *Backend_API.Tests* for the 99-test suite. The live smoke scripts run against `http://localhost:5050` with the seeded demo accounts. Document 3 gives the human step-by-step version of every scenario.